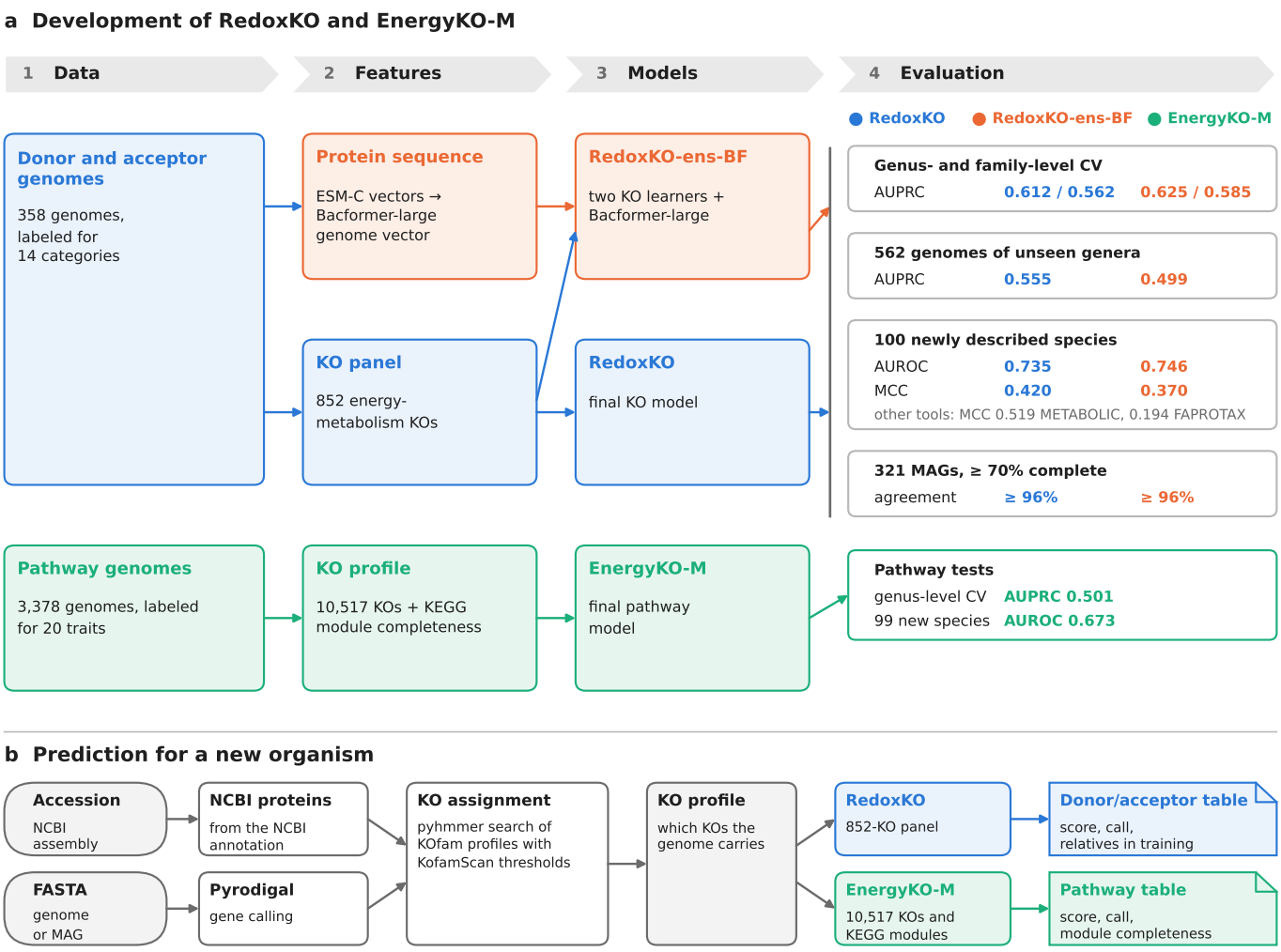
a Development of RedoxKO and EnergyKO-M
1 Data
2 Features
3 Models
4 Evaluation
RedoxKO
RedoxKO-ens-BF
EnergyKO-M
Donor and acceptor genomes
358 genomes, labeled for 14 categories
Protein sequence
ESM-C vectors → Bacformer-large genome vector
RedoxKO-ens-BF
two KO learners + Bacformer-large
KO panel
852 energy-metabolism KOs
RedoxKO
final KO model
Genus- and family-level CV
AUPRC 0.612 / 0.562 0.625 / 0.585
562 genomes of unseen genera
AUPRC 0.555 0.499
100 newly described species
AUROC 0.735 0.746
MCC 0.420 0.370
other tools: MCC 0.519 METABOLIC, 0.194 FAPROTAX
321 MAGs, ≥ 70% complete
agreement ≥ 96% ≥ 96%
Pathway genomes
3,378 genomes, labeled for 20 traits
KO profile
10,517 KOs + KEGG module completeness
EnergyKO-M
final pathway model
Pathway tests
genus-level CV AUPRC 0.501
99 new species AUROC 0.673
b Prediction for a new organism
Accession
NCBI assembly
FASTA
genome or MAG
NCBI proteins
from the NCBI annotation
Pyrodigal
gene calling
KO assignment
pyhmmer search of KOfam profiles with KofamScan thresholds
KO profile
which KOs the genome carries
RedoxKO
852-KO panel
EnergyKO-M
10,517 KOs and KEGG modules
Donor/acceptor table
score, call, relatives in training
Pathway table
score, call, module completeness